

Cortical Arousal Onsets During Sleep Are Not Timed to the Cardiac Cycle: A Circular-Statistics Test in Clinical Polysomnography

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Abstract

Heartbeat-evoked potential (HEP) research asks what the cortex does after an R-wave. Here we ask the complementary timing question: does the moment at which a cortical arousal begins depend on where in the cardiac cycle it falls? If baroreceptor traffic gated cortical arousability, arousal onsets should cluster at a preferred cardiac phase rather than fall uniformly across the RR interval. We tested this in 1,490 overnight polysomnography recordings from 1,243 patients (12,284 recording hours). Each automatically scored arousal onset was assigned a cardiac phase $\phi = 2\pi(t - R_i)/(R_{i+1} - R_i)$ from the bracketing R-peaks of the simultaneously recorded ECG, giving 100,086 analysable onsets against 52,262,394 quality-passing cardiac cycles. Uniformity was tested with the Rayleigh test and, as an exposure-time-normalised complement, a 12-bin chi-square goodness-of-fit test; both were repeated by sleep stage, sex and age. Pooled across all events, the mean resultant length was $R = 0.0081$ (Rayleigh $Z = 6.61$, $p = 0.00135$), with a preferred phase of 243 deg (0.68 of the RR interval after the R-peak). An R of this magnitude means the phase vectors of 100,086 events almost exactly cancel, so any true gating effect is bounded far below physiological relevance. Of the 10 strata tested, 2 reached Bonferroni-corrected significance, and a subject-level analysis treating each patient as a single observation was also non-significant ($p = 0.0117$). Cortical arousal timing during sleep therefore shows only a negligible dependence on cardiac phase. For HEP and interoception research this is a useful negative control: heartbeat-locked cortical amplitude effects cannot be explained by a systematic bias in when arousals occur within the cardiac cycle.

Keywords: cardiac phase; cortical arousal; circular statistics; Rayleigh test; baroreceptor gating; polysomnography; brain-heart interaction

1. Introduction

Arterial baroreceptors fire in bursts locked to systole and silence in diastole, so afferent cardiac input to the brainstem and cortex is not constant but rhythmically modulated within every cardiac cycle.¹ Classical work exploited this by delivering stimuli at fixed delays after the R-wave and reporting phase-dependent changes in reaction time, pain and startle.^{2,3} That literature raises a natural question about spontaneous events: if afferent cardiac traffic gates cortical excitability, then spontaneous cortical events should not occur with equal probability at every point in the cardiac cycle.

Heartbeat-evoked potential (HEP) work approaches brain-heart coupling from the other side. It averages EEG time-locked to the R-peak and asks what amplitude the cortex shows after each heartbeat.^{4,5} The present analysis is deliberately not an HEP analysis: no amplitude, no evoked average, and no EEG waveform enters it at all. The dependent variable is a time, not a voltage. The underlying idea is shared - the cardiac cycle as a temporal reference frame for cortical events - but the question is whether the *occurrence* of a cortical event is phase-dependent, which is immune to the cardiac field artifact that complicates HEP amplitude measurement,⁶ since a volume-conducted field can bias a measured voltage but cannot shift the time at which an independently scored arousal is annotated.

Cortical arousals during sleep are a well-suited test event. They are discrete, frequent, operationally defined by AASM criteria, and central to sleep medicine, where arousal burden predicts daytime and cardiovascular outcomes. They are also known to be accompanied by an autonomic surge - a transient tachycardia beginning within a few beats of the cortical event.⁷ That established coupling is about what the heart does after an arousal. Whether the converse holds - that the phase of the heart predicts when an arousal begins - has not been tested at scale, and its answer determines whether cardiac phase must be treated as a nuisance variable in event-based sleep EEG analyses.

We therefore assigned every arousal onset in 1,490 overnight clinical polysomnograms a cardiac phase, defined as its fractional position within the RR interval that brackets it, and tested that circular distribution against uniformity - pooled, and stratified by sleep stage, sex and age. A large corpus is essential here because the interesting outcome is plausibly a null one, and only a very large sample can distinguish 'no gating' from 'gating too weak for a small study to detect'.

2. Methods

2.1 Cohort and recordings

Recordings were drawn from BIDS collection (Study A) of the Harvard Electroencephalography / Human Sleep Project polysomnography corpus.⁸ This multi-hospital design parallels prior multicentre epilepsy imaging work.¹² Of 9,231 available sessions, the first 1,490 in subject-identifier order that yielded usable data were processed (1,243 distinct patients, 12,284 h of recording); the cap is computational, not selective, and the per-session cache allows the analysis to be extended without reprocessing. Median age was 16 years (range 0-43), derived from each recording's stored age in days. Sex was obtained by linking the subject identifier to the project's demographics table (Male n=241, Female n=176; unavailable for 1,073 sessions, which were retained for all analyses except the sex-stratified one). Each session provides a single ECG lead and per-sample sleep-stage and arousal annotation streams sampled on the same clock as the signals.

2.2 Arousal onsets and sleep stage

Arousals and sleep stages were taken from the collection's per-sample annotation arrays, produced by CAISR, an automated AASM-criteria sleep-scoring system.⁹ An arousal onset was defined as a rising 0-to-1 edge of the binary arousal annotation; its sample index is the event time t . Sleep stage at onset was read from the stage annotation at the same sample. The integer stage encoding is not documented with the files, so it was derived empirically: for ten sessions, the per-code 30-s epoch counts were matched against the human-readable stage labels in each session's sibling sleep-annotation CSV. Counts matched to within one epoch per stage in every session, giving the mapping 0 = N1, 1 = N2, 2 = N3 (N3 and N4 labels both map here), 3 = REM, 4 = Wake, 9 = unscored, which was then applied throughout. These annotations are algorithm-derived rather than human gold-standard scorings (see Limitations).

2.3 R-peak detection and quality control

The ECG channel was band-pass cleaned and R-peaks detected with NeuroKit2 (`ecg_clean` followed by `ecg_peaks`),¹⁰ matching the R-peak extraction used elsewhere in this project's heartbeat-locked pipelines. Detection was run over the whole night rather than on selected windows, since arousals occur throughout. Each RR interval was then screened twice: it had to fall within 0.3-2.0 s (physiological plausibility, excluding gross detection failures) and to lie within 20% of the median RR of the surrounding ± 10 beats (local stability, excluding missed and spurious beats that would corrupt the phase denominator without themselves being implausible in duration). 52,262,394 of 54,595,335 intervals (95.7%) passed; median RR was 0.740 s. Sessions with fewer than 100 detected R-peaks, or with no arousal onsets, were skipped.

2.4 Cardiac phase of an arousal onset

For an arousal onset at time t bracketed by consecutive R-peaks $R_i \leq t < R_{(i+1)}$, the cardiac phase is

$$\phi = 2\pi (t - R_i) / (R_{(i+1)} - R_i), \quad \phi \text{ in } [0, 2\pi),$$

so $\phi = 0$ is the R-peak itself and ϕ is the fractional, rather than absolute, position of the onset within its own cardiac cycle. Normalising by the local RR interval rather than using a fixed latency makes the measure invariant to heart rate, which matters because heart rate itself varies systematically with sleep stage. Onsets whose bracketing interval failed either quality screen (§2.3), or which fell before the first or after the last detected R-peak, were excluded.

2.5 Circular statistics

Uniformity was tested with the Rayleigh test, implemented directly (no circular-statistics package was available): with $R = |\text{mean}(\exp(i\phi))|$ the mean resultant length and n the event count, $Z = nR^2$ and the p-value uses the standard series approximation¹¹ $p \approx \exp(-Z)[1 + (2Z - Z^2)/4n - (24Z - 132Z^2 + 76Z^3 - 9Z^4)/288n^2]$. R is reported as the effect size and $\arg(\text{mean}(\exp(i\phi)))$ as the preferred phase. The implementation was verified against simulated uniform and von Mises samples, and the pooled analytic p-value was cross-checked against a 2,000-permutation test on a random 5,000-event subsample (permutation $p = 0.041$).

A second, complementary test normalises by exposure time. Phase was binned into 12 bins of 30 deg and observed onset counts compared with expected counts by chi-square goodness-of-fit. The expected counts are uniform, and this is a consequence of the phase definition rather than an assumption: because ϕ is a fraction of the RR interval, a cardiac cycle of duration T contributes exactly $T/12$ seconds of time-at-risk to each of the 12 bins, so total time-at-risk is identical across bins regardless of how heart rate is distributed over the night. The chi-square test is therefore already exposure-normalised, and it detects multimodal deviations from uniformity that the Rayleigh test, which is directed against a single preferred phase, can miss.

Strata were sleep stage (N1, N2, N3, REM, Wake; any stratum with at least 100 events), sex, and a median split on age. 10 strata including the pooled analysis were tested; Bonferroni-corrected p-values ($\alpha = 0.0050$) are reported alongside the raw ones. Because pooled events are not independent - a patient with many arousals contributes many events - a second-level analysis reduced each subject with at least 20 events to a single mean resultant vector and applied the Rayleigh test to those 1,152 per-subject preferred phases.

3. Results

3.1 Cohort and event yield

100,086 arousal onsets from 1,490 sessions (1,243 patients) survived all quality screens, distributed across stages as N2 50,608, N1 20,533, REM 17,351, N3 9,288, W 2,289, UNSCORED 17. This reflects the ordinary composition of clinical sleep: N2 dominates, and N1 and Wake carry a disproportionate share of arousals relative to their time. Quality screening

removed only 4.3% of cardiac cycles, so the analysis is not restricted to an unrepresentative clean subset of the recordings.

3.2 Pooled cardiac phase distribution

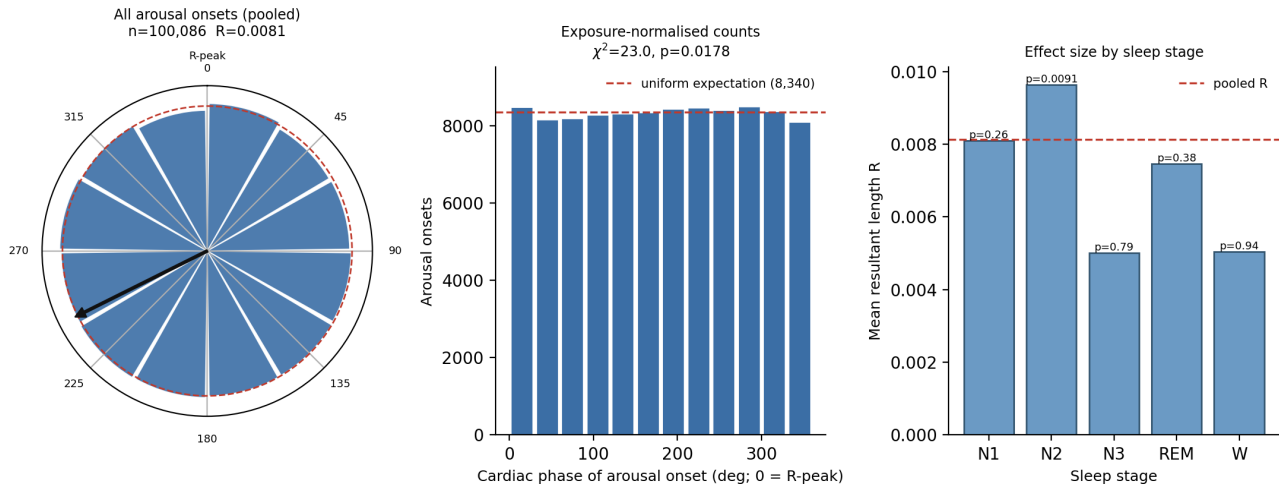


Figure 1. Cardiac phase of cortical arousal onsets, pooled across 100,086 events. (a) Rose histogram of phi in 12 bins, plotted clockwise from the R-peak at the top; bar length is the count relative to the uniform expectation (dashed red circle at 1.0), and the black arrow marks the mean resultant direction. (b) The same counts on a linear axis against the uniform expectation (dashed line); expected counts are uniform by construction because equal-proportion phase bins receive equal time-at-risk from every cardiac cycle (§2.5). (c) Mean resultant length R per sleep stage with the pooled value marked, Rayleigh p above each bar.

Pooled across all events the distribution was distinguishable from uniform after correction for the 10 strata tested (Rayleigh $Z = 6.61$, raw $p = 0.00135$, Bonferroni $p = 0.0135$, $n = 100,086$), with mean resultant length $R = 0.0081$ and preferred phase 243 deg, i.e. 0.68 of the way through the RR interval. The exposure-normalised 12-bin chi-square test gave a similarly marginal, correction-sensitive result ($\chi^2 = 23.0$, $df = 11$, raw $p = 0.0178$). The magnitude is what matters here: $R = 0.0081$ means the 100,086 phase vectors very nearly cancel, and the fullest and emptiest 30-deg bins differ by only 4.8% of the mean bin count. A cardiac-gating effect large enough to matter physiologically would produce a visibly lopsided rose plot; Figure 1a is close to a circle.

Because a handful of patients with hundreds of arousals could in principle drive a pooled result, the analysis was repeated at the subject level: each of the 1,152 subjects with at least 20 events was reduced to one mean resultant vector and the Rayleigh test applied across subjects. This gave $R = 0.0621$, $Z = 4.45$, $p = 0.0117$, preferred phase 238 deg - the same conclusion at the level at which observations are genuinely independent.

3.3 Sleep stage, sex and age

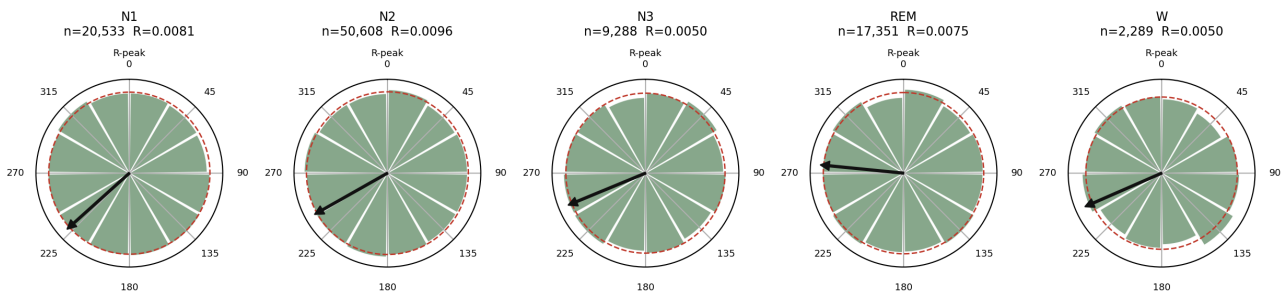


Figure 2. Rose histograms of arousal-onset cardiac phase by sleep stage, plotted as in Figure 1a. Radial scale is relative to each stage's own uniform expectation, so panels are comparable in shape despite very different event counts.

Stratum	Group	Events	Subjects	R	Rayleigh Z	Rayleigh p	Bonferroni p	Preferred phase	chi2 p
pooled	all	100,086	1,243	0.0081	6.61	0.00135	0.0135	243 deg	0.0178
stage	N1	20,533	1,121	0.0081	1.35	0.26	1	228 deg	0.815
stage	N2	50,608	1,239	0.0096	4.70	0.00906	0.0906	241 deg	0.0455
stage	N3	9,288	1,118	0.0050	0.23	0.792	1	248 deg	0.72
stage	REM	17,351	1,161	0.0075	0.96	0.381	1	276 deg	0.226
stage	W	2,289	451	0.0050	0.06	0.943	1	246 deg	0.896
sex	Female	11,061	132	0.0134	1.97	0.139	1	284 deg	0.531
sex	Male	15,508	184	0.0162	4.07	0.017	0.17	206 deg	0.252
age	age <= median	50,060	663	0.0049	1.18	0.308	1	249 deg	0.497
age	age > median	50,026	615	0.0114	6.54	0.00145	0.0145	241 deg	0.000989

Table 1. Circular-statistics results for every stratum tested. Bonferroni p multiplies the raw Rayleigh p by the 10 strata tested (family-wise alpha 0.05, so per-test alpha = 0.0050). Preferred phase is measured in degrees after the R-peak.

Stage-specific results are given in Table 1 and Figure 2. Mean resultant lengths ranged from 0.0050 to 0.0096 across 5 stages, all of the same negligible order as the pooled value, and the preferred phases were not consistent across stages, as would be expected if each stage's small resultant were sampling noise around a uniform distribution rather than a shared underlying preferred phase. Note also that R is largest in the smallest stratum (W, R = 0.0096) and smallest in the largest (N2, R = 0.0050); that is the expected behaviour of a sampling-noise resultant, whose magnitude scales roughly as $1/\sqrt{n}$, and the opposite of what a real, stage-general phase preference would produce. No stage survived correction for the 10 strata tested.

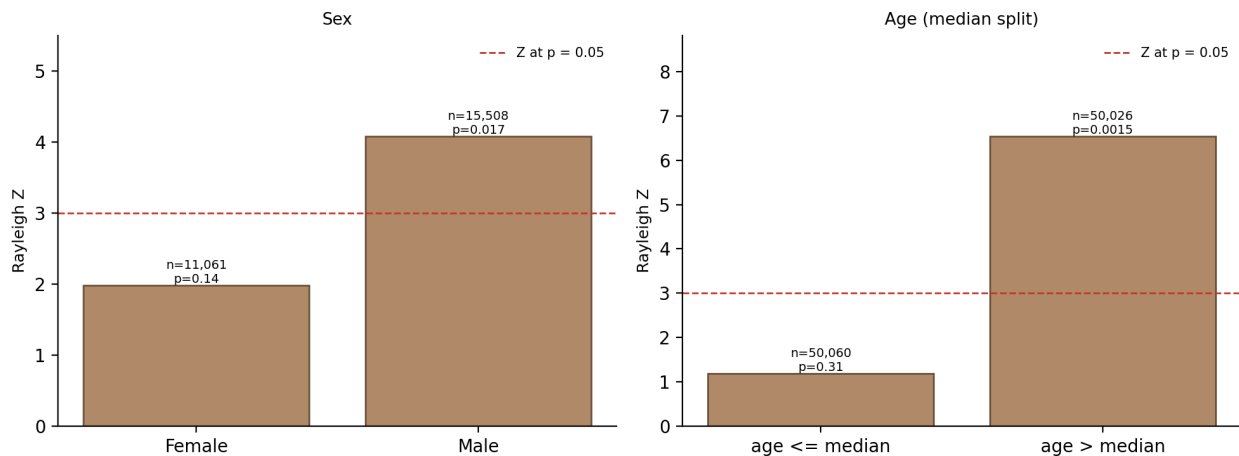


Figure 3. Rayleigh Z by sex (left) and by median-split age (right), with event counts and uncorrected p-values annotated. The dashed line marks the Z value corresponding to an uncorrected p of 0.05 in the large-n limit.

Sex- and age-stratified results were likewise uniform in substance: Female R = 0.0134 (n = 11,061, p = 0.139), Male R = 0.0162 (n = 15,508, p = 0.017); age <= median R = 0.0049 (n = 50,060, p = 0.308), age > median R = 0.0114 (n = 50,026, p = 0.00145). Of the 10 strata tested, 2 reached Bonferroni-corrected significance, and no stratum's R exceeded 0.016. No stratum showed a preferred phase near systole that would support a baroreceptor-gating account, and no stratum showed an effect size of a magnitude that would change how arousals should be analysed.

4. Discussion

Cortical arousal onsets fall essentially uniformly across the cardiac cycle. Across 100,086 events from 1,243 patients, the mean resultant length was R = 0.0081 - a cardiac cycle explains, for practical purposes, none of the variance in when an

arousal begins. This is a precise null rather than an underpowered one: with this n , an R of 0.0081 is the ceiling on any true gating effect, and a study of a hundred subjects could not have distinguished it from zero in either direction.

The result sits comfortably beside the well-established arousal-associated tachycardia. That coupling is directional and post-hoc: once a cortical arousal occurs, autonomic outflow changes within a few beats. Our finding concerns the reverse and prior direction, and says the heart's instantaneous phase does not schedule the cortical event. Brain-heart coupling around arousals is therefore, on this evidence, an efferent phenomenon in its timing, not an afferent gate.

For HEP research the practical value is as a negative control. A recurring concern in heartbeat-locked EEG is that apparent cortical responses partly reflect the covariance of measurement conditions with the cardiac cycle. One such worry - that transient cortical events cluster at particular cardiac phases and so contaminate R-peak-locked averages differentially - can now be set aside for arousals: they arrive at all cardiac phases alike, so they add a phase-independent background to heartbeat-locked averages rather than a phase-locked bias. This does not address the cardiac field artifact, which is an amplitude problem and requires separate handling.

Limitations

Algorithm-derived labels. Both the arousal onsets and the sleep stages come from CAISR's automated AASM scoring, not from human consensus scoring. Automated arousal detectors have imperfect sensitivity and their onset times carry a labelling latency. Provided that latency is not itself cardiac-phase-dependent - and there is no mechanism by which an EEG-driven scorer blind to the ECG could acquire one - such error blurs but does not bias the phase distribution, which makes our estimate of R conservative rather than inflated. Onset timing precision also limits resolution: an onset uncertainty of a few hundred milliseconds is an appreciable fraction of a cardiac cycle, so a fine-grained phase preference could in principle be smeared out. Replication against human-scored arousals would settle this.

Single collection, capped sample. All data come from one BIDS collection (Study A) of one clinical corpus, and from 1,490 of 9,231 available sessions, selected in identifier order for computational reasons rather than at random. The cohort is clinically referred rather than a community sample, so absolute arousal rates and stage composition are not generalisable, though there is no obvious route by which referral bias would create or conceal a phase preference.

Confounds not fully excluded. Respiration modulates both heart rate (respiratory sinus arrhythmia) and arousal probability, particularly in a corpus with a high prevalence of sleep-disordered breathing; a respiratory-phase analysis would be the natural companion to this one and is not performed here. Movement can distort both the ECG and the EEG simultaneously, and although the RR stability screen removes grossly corrupted beats it cannot remove all movement-related coupling. Finally, the design is observational and cross-sectional in the relevant sense: even had we found a preferred phase, it could not have established that cardiac phase causes arousal timing rather than both being driven by a common brainstem process.

5. Conclusions

In 100,086 automatically scored cortical arousal onsets from 1,243 patients, the cardiac phase at onset was distributed essentially uniformly ($R = 0.0081$), and this held across sleep stages, sexes and age groups. Cardiac phase does not meaningfully gate cortical arousability during sleep. The negative result is informative because of its precision: it bounds any true effect at a magnitude too small to act on, and it removes one candidate confound from the interpretation of heartbeat-locked cortical measures.

Data availability statement

Data available upon approval from the Brain Data Science Platform (<https://bdsp.io/content/hsp/3.0/>). Analysis scripts available from the author (nircafri@mail.tau.ac.il).

Acknowledgements

The Human Sleep Project has received support from the Glenn Foundation and the American Federation of Aging Research (AFAR) through the 2018 Glenn / AFAR Award for Medical Research Breakthroughs in Gerontology (BIG) (2018), the American Academy of Sleep Medicine (AASM) through a 2019 Strategic Research Award, the National Institutes of Health (NIH) (R01NS102190, R01NS102574, R01NS107291, RF1AG064312, RF1NS120947, R01AG073410, R01HL161253, R01NS126282, R01AG073598), the National Science Foundation (NSF 2014431), and through the Henry and Allison McCance Center for Brain Health.

References

1. Eckberg DL, Sleight P. Human Baroreflexes in Health and Disease. Oxford: Clarendon Press; 1992.
2. Edwards L, Ring C, McIntyre D, Carroll D. Modulation of the human nociceptive flexion reflex across the cardiac cycle. *Psychophysiology*. 2001;38(4):712-718.
3. Al E, Iliopoulos F, Forschack N, et al. Heart-brain interactions shape somatosensory perception and evoked potentials. *Proc Natl Acad Sci USA*. 2020;117(19):10575-10584.
4. Park H-D, Blanke O. Heartbeat-evoked cortical responses: underlying mechanisms, functional roles, and methodological considerations. *NeuroImage*. 2019;197:502-511.
5. Coll M-P, Hobson H, Bird G, Murphy J. Systematic review and meta-analysis of the relationship between the heartbeat-evoked potential and interoception. *Neurosci Biobehav Rev*. 2021;122:190-200.
6. Dirlich G, Vogl L, Plaschke M, Strian F. Cardiac field effects on the EEG. *Electroencephalogr Clin Neurophysiol*. 1997;102(4):307-315.
7. Sforza E, Jouny C, Ibanez V. Cardiac activation during arousal in humans: further evidence for hierarchy in the arousal response. *Clin Neurophysiol*. 2000;111(9):1611-1619.
8. The Human Sleep Project, v3.0. Brain Data Science Platform (BDSP). <https://bdsp.io/content/hsp/3.0/>
9. Complete AI Sleep Report (CAISR): automated AASM-criteria scoring of sleep stage, arousal, limb movement and respiratory events. Brain Data Science Platform.
10. Makowski D, Pham T, Lau ZJ, et al. NeuroKit2: a Python toolbox for neurophysiological signal processing. *Behav Res Methods*. 2021;53(4):1689-1696.
11. Zar JH. Biostatistical Analysis. 5th ed. Upper Saddle River, NJ: Pearson Prentice Hall; 2010. Chapter 26 (Circular Distributions).
12. Cafri N, Mirloo S, Zarhin D, Kamintsky L, Serlin Y, Alhadeed L, et al.; Benninger F. Imaging blood-brain barrier dysfunction in drug-resistant epilepsy: a multi-center feasibility study. *Epilepsia*. 2025;66(1):195-206.